Predicting Sleep Apnea Severity Based on Physiological and Demographic Data

Group 5: Lydia Tembreull

SAT 5114: AI in Healthcare

Problem Statement

Condition: Obstructive Sleep Apnea (OSA) involves the repeated collapse of the upper airway during sleep, leading to oxygen desaturation and sleep disturbance

Problem: OSA is a global epidemic linked to obesity, cardiovascular disease, and stroke, but it remains severely under-diagnosed

Current Limitations: The “gold standard” diagnosis, Polysomnography (PSG), is costly, labor-intensive, and manually scored, causing significant delays in patient care

AI Solution: Automated Clinical Decision Support can leverage existing demographic and physiological data to prioritize high-risk patients for screening

Literature Review

Evolution of AI in Sleep Medicine: Research has shifted from simple accuracy-driven models to “Explainable AI” (XAI) to help clinicians understand decision-making

Key Findings: Studies using the Sleep Heart Health Study (SHHS) dataset have shown that heart rate variability and oxygen metrics can predict cardiovascular mortality, not just apnea

AI Methodologies: Past works have successfully used algorithms like Random Forest and XGBoost to categorize severity, with accuracy benchmarks typically ranging from 67% to 88% and AUC benchmarks from 0.69 to 0.98

- Ex: While Shi et al. (2023) achieved 75.2% accuracy and an AUC score of 0.840 for severe cases using XGBoost, my study explores if Logistic Regression can offer comparable screening utility with higher interpretability

Data & Population Justification

Dataset: Sleep Heart Health Study (SHHS) Visit 1 (5,804 patient records)

Population Selection: A diverse multi-center cohort funded by the NHLBI, ensuring high-quality, standardized PSG data

Justification: Features like Age, BMI, and Neck Circumference were chosen as primary predictors due to their established clinical correlation with airway collapsibility

Target Classes: Patients categorized by Apnea-Hypopnea (AHI):

- None (< 5)
- Mild (5-15)
- Moderate (15-30)
- Severe (≥ 30)

Dataset Strengths & Weaknesses

Strengths:

- **High Quality**: Standardized equipment and centralized “Reading Center” for reliability
- **Large Sample**: over 5,000 records allow for robust training and validation

Weaknesses:

- **Class Imbalance**: “Severe” is under-represented (681 samples vs. 2,041 for “Mild”)
- **Missing Data**: Significant missing values in features like average heart rate (avhrbp) required heavy imputation

Methodology & Code Structure

Preprocessing:

- Two-phase imputation (Median for continuous, Mode for categorical)
- One-Hot Encoding for gender, race, and smoking status; StandardScaler for numerical features

Algorithms Compared: Logistic Regression, Random Forest, XGBoost, and SVM

Validation: 70/15/15 split (Train/Validation/Test) with 5-fold cross-validation

Explainability: SHAP (Shapley Additive exPlanations) used to visualize feature importance

Working Code Demo & Analysis

Conclusion & Recommendations

Conclusion: Model is stable and generalizes well. The 53.6% overall accuracy reflects the clinical difficulty distinguishing contiguous classes (Mild vs. Moderate), but the 0.88 AUC proves the model is highly effective at the binary task of 'Severe vs. Not Severe,' which is the most critical screening goal. Discovered that male gender, age, neck circumference, and BMI are the strongest positive predictors of OSA, while oxygen saturation is the strongest negative predictor.

Recommendations:

1. **Interaction Terms**: Expand on BMI x Age interactions to capture non-linear aging effects
2. **Algorithm Optimization**: Refine ensemble methods like Random Forest and XGBoost more aggressively to capture non-linearities
3. **Feature Expansion**: Incorporate Oxygen Desaturation Index (ODI) and Nightly SPO_2 fluctuations to help the model better distinguish "Mild" from "Moderate" cases

Feasibility: Recommendations are highly feasible because the data required use existing clinical data and standard libraries, making them easy to implement in a hospital setting

References

Goldstein, C. A., et al. (2020). Artificial Intelligence in sleep medicine: Background and implications for clinicians. *Journal of Clinical Sleep Medicine.*

Li, S., et al. (2025). Advances in Machine Learning Prediction Models for the Screening of Obstructive Sleep Apnea in Adults. *Nature and Science of Sleep.*

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *31st Conference on Neural Information Processing Systems (NIPS).*

Maniaci, A., et al. (2023). Machine Learning Identification of Obstructive Sleep Apnea Severity through the Patient Clinical Features. *MDPI.*

National Sleep Research Resources (2026). *Sleep Heart Health Study (SHHS) Visit 1 Dataset.* [https://sleepdata.org/datasets/shhs].

Punjabi, N. M. (2008). The epidemiology of adult obstructive sleep apnea. *Proceedings of the American Thoracic Society.*

Quan, S. F., et al. (1997). The sleep heart healthy study: Design, rationale, and methods. *Sleep.*

Shi, Y., et al. (2023). Application and interpretation of machine learning models in predicting the risk of severe obstructive sleep apnea in adults. *BMC Medical Information and Decision Making.*